

Practice problems #8

1. For the following matrices, give the angle θ of the rotation, where $-\pi < \theta \leq \pi$, and give the scale factor γ .

(a)

$$\begin{bmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{bmatrix}$$

(b)

$$\begin{bmatrix} 0 & 0.3 \\ -0.3 & 0 \end{bmatrix}$$

2. True or False. Justify:

Let A be a real $n \times n$ matrix and let $\mathbf{v}$ in $\mathbf{C}^n$ be an eigenvector corresponding to a complex eigenvalue λ . For each nonzero complex scalar μ , the vector $\mu\mathbf{v}$ is also an eigenvector of A .

3. Find an invertible matrix P and a matrix C of the form $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ such that the given matrix has the form $A = PCP^{-1}$.

$$A = \begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix}$$

4. Justify the following statement *without* any calculation:

If a and b are real and $A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, then the vectors $\begin{bmatrix} 1 \\ -i \end{bmatrix}$ and $\begin{bmatrix} 1 \\ i \end{bmatrix}$ must be eigenvectors of A .

Hints or Answers on page 2.

#1(a): $\theta = \pi/6$, $\gamma = 2$

#1(b): $\theta = -\pi/2$, $\gamma = 0.3$

#2: True. Justify!

#3: Once you find an eigenvalue of A , and its corresponding eigenvector v , you can fill out columns of P and the elements of C , right? There can be many answers because there are infinitely many eigenvectors. One example is: $P = \begin{bmatrix} 1 & 3 \\ 2 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 2 & -3 \\ 3 & 2 \end{bmatrix}$

#4: Just by looking at A , you can find the eigenvalues of A , yes? Note that $P = I$ in this case. Are the eigenvectors are correct to be columns of the identity matrix that is P ?